Name: ____________________

FRACTIONS, DECIMALS AND PERCENTAGES
(CONVERTING BETWEEN FDP)

Date: ________________

LO: I can convert fluently between fractions, decimals and percentages.

What does converting mean?

Converting means writing a number in a **DIFFERENT** form without changing its value. Fractions, decimals and percentages are three ways to write the same amount.

Key fact: To convert a fraction to a decimal, divide the numerator by the denominator. To convert a decimal to a percentage, multiply by 100.

Equivalent values

🟢	½ = 0.5 = 50%	all equivalent
🟢	¼ = 0.25 = 25%	all equivalent
🟢	⅗ = 0.6 = 60%	all equivalent
🔴	⅓ = 0.3 = 30%	0.3 ≠ ⅓
🔴	0.7 = 70% = $\frac{7}{70}$	should be $\frac{7}{10}$

Key idea

Fraction Decimal: divide top by bottom. Decimal Percentage: × 100. Percentage Fraction: write over 100 and simplify.

⅜ 3 ÷ 8 = 0.375 37.5%

Divide, then multiply by 100

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – fraction to decimal

Convert ¾ to a decimal

3 ÷ 4

= 0.75

= 0.75

Divide numerator by denominator: 3 ÷ 4 = 0.75

Example 2 – decimal to percentage

Convert 0.35 to a percentage

0.35 × 100

= 35

Add % sign

= 35%

Multiply the decimal by 100 and add the % symbol.

Example 3 – percentage to fraction

Convert 45% to a fraction in simplest form

45/100

HCF of 45 and 100 = 5

= 9/20

= $\frac{9}{20}$

Write over 100, then simplify by dividing both by the HCF (5).

Example 4 – fraction to percentage

Convert $\frac{7}{20}$ to a percentage

7 ÷ 20 = 0.35

0.35 × 100

= 35%

= 35%

Convert to decimal first (divide), then multiply by 100.

YOUR TURN – PRACTICE (deliberate practice)

1. Fraction to decimal

Convert each fraction to a decimal by dividing the numerator by the denominator.

a) $\frac{1}{2}$

b) $\frac{1}{5}$

c) $\frac{3}{8}$

d) $\frac{7}{20}$

2. Decimal to percentage

Convert each decimal to a percentage by multiplying by 100.

a) 0.4

b) 0.75

c) 0.08

d) 0.125

3. Percentage to fraction

Convert each percentage to a fraction in its simplest form. Write over 100 then simplify.

a) 20%

b) 60%

c) 35%

d) 84%

4. Fraction to percentage

Convert each fraction to a percentage. Divide first, then multiply by 100.

a) $\frac{1}{4}$

b) $\frac{2}{5}$

c) $\frac{3}{8}$

d) $\frac{11}{20}$

5. Mixed conversions

Convert each value to the other two forms (fraction, decimal, percentage).

a) 0.6 = ___ = ___%

b) 72% = ___ = ___

c) $\frac{9}{25}$ = ___ = ___%

d) 0.875 = ___ = ___%

e) 5% = ___ = ___

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) $\frac{1}{3}$ **= 0.3** ☐	**b) 0.25 = 25%** ☐	**c) 70% =** $\frac{7}{10}$ ☐	**d) 0.5% =** $\frac{1}{2}$ ☐	**e)** $\frac{4}{5}$ **= 0.45** ☐
Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________	Correct answer: __________
Reason: ________________	Reason: ________________	Reason: ________________	Reason: ________________	Reason: ________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A shop offers a $\frac{1}{4}$ off sale. The sale sign says "20% off everything!"
Is the sale sign correct? Show your working to explain why or why not.

Expression: ____________________________

Simplified: ____________________________

b) Three students scored the following on different tests:
Amy: 17 out of 20   Ben: 0.82   Clara: 78%
Who got the highest score? Convert all to the same form to compare.

Expression: ____________________________

Simplified: ____________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!